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Project 2

Comparative Study of Image Classification Using Decision Tree, Naive Bayes, and Feedforward Neural Networks

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Introduction

Image classification is a core task in the field of artificial intelligence and machine learning, especially when applied to medical data. The ability to automatically identify and categorize medical images has the potential to assist doctors in diagnosing diseases more accurately and efficiently. In this project, we focus on classifying brain tumor images using three different supervised learning models: Naive Bayes, Decision Tree, and Multi-Layer Perceptron MLP. Each model represents a different learning paradigm: probabilistic reasoning, rule-based logic, and neural networks.

The main objective of this study is to compare the performance of these models on the same dataset and evaluate them based on their ability to learn meaningful patterns from image data. We explore the strengths and weaknesses of each classifier, not only in terms of accuracy, but also in terms of training complexity, interpretability, and scalability. The final goal is to determine which model is more suitable for real-world medical image classification.

Dataset Description

The dataset used in this project is a labeled collection of brain MRI images, organized into three main categories: glioma tumor, pituitary tumor, and no tumor. Each category is represented by a folder containing multiple images in standard formats such as JPG and PNG. The dataset includes more than 2000 images in total, with relatively balanced class distributions.

To prepare the data for machine learning, we applied a series of preprocessing steps. All images were resized to a fixed resolution of 64×64 pixels to ensure consistency across samples. Each image was then flattened into a one-dimensional array of pixel values, forming the feature vector used for classification. The class labels were also encoded numerically using label encoding. Finally, we split the dataset into training and testing sets using an 80/20 ratio. For the MLP model specifically, we applied feature scaling to normalize the input data before training. These preprocessing steps helped ensure that all models received the data in a format suitable for learning.

Implemented Models

Naive Bayes Classifier

The Naive Bayes classifier is a probabilistic model that relies on Bayes' Theorem to make predictions. It assumes that all features are conditionally independent given the class label, which simplifies the learning process and allows the model to scale well to high-dimensional data.

In the context of classification, Naive Bayes calculates the posterior probability for each class and selects the one with the highest value. It is particularly known for its fast-training speed and effectiveness with large datasets, even when the independence assumption is not fully satisfied. The Gaussian variant of Naive Bayes is commonly used when features are continuous and assumed to follow a normal distribution.

In our project, we implemented the Naive Bayes classifier using the `GaussianNB()` function from the Scikit-learn library. Each image, represented as a flattened vector of 4096-pixel values, was treated as a feature vector input to the model. Despite the unrealistic independence assumption (since image pixels are clearly correlated), the model still produced a reasonable baseline performance.

The classifier was trained on the training portion of the dataset and then tested on the remaining samples. The results, including accuracy, precision, recall, and F1-score, were used as reference to compare with the more advanced models.

Decision Tree Classifier

A Decision Tree is a rule-based classifier that organizes decisions in a hierarchical tree structure. At each internal node, the model selects the most informative feature and splits the data based on its values. This process continues recursively until the data is either perfectly separated or a stopping condition is reached. The model uses information gain, often calculated using entropy, to determine which features to split on.

Decision Trees are appreciated for their interpretability, as they can be visualized and understood as a sequence of if-else rules. However, they are prone to overfitting, especially when the tree becomes too deep.

In the context of our project, we used Scikit-learn's `DecisionTreeClassifier` with entropy as the splitting criterion. After training on the tumor image dataset, we visualized the top three levels of the decision tree using `plot_tree()`. This allowed us to inspect how the model made decisions based on the pixel values. Even though the input space is large (4096 dimensions), the tree was able to learn some structure from the data and achieved reasonable performance. However, the risk of overfitting remained due to the complexity and high dimensionality of the input.

Multi-Layer Perceptron MLP (Feedforward Neural Network)

The Multi-Layer Perceptron MLP is a type of feedforward artificial neural network composed of multiple layers of interconnected neurons. Each neuron performs a weighted sum of its inputs followed by an activation function such as ReLU.

The MLP learns by adjusting its weights using a process called backpropagation, where the error is propagated backward through the network to update the weights using gradient descent. Unlike simple perceptrons, MLPs can model non-linear relationships and are capable of learning complex patterns in the data. They are especially powerful for tasks like image recognition where features are not linearly separable.

In our implementation, we used the `MLPClassifier` from Scikit-learn with two hidden layers containing 256 and 128 neurons respectively. Before training, we normalized the data using `StandardScaler` to improve convergence speed and stability. The model was trained for up to 500 iterations using backpropagation. Due to its high capacity, the MLP performed well in identifying tumor types, capturing subtle differences in image patterns. However, it required more computational resources and time compared to the other two models. It also lacked the interpretability of Decision Trees, but its accuracy made it a strong candidate for real-world medical applications.

Model Evaluation: Accuracy Metrics and Confusion Matrices

Accuracy, precision, recall, and F1-score are common classification metrics that we used to assess all three models. Evaluation functions were used to calculate these metrics.

The confusion matrices that follow show how well each model performed in dividing the brain MRI pictures into three groups: pituitary tumor, glioma tumor, and no tumor. Each column shows the class that the model predicted, and each row shows the actual (true) class.

Correct classifications are represented by values along the diagonal, while incorrect classifications are represented by values off the diagonal. Such matrices allow to perceive which classes are being misled by the models and provide an indication of their strengths and weaknesses regarding image-based tumor detection.

Naive Bayes Classifier

The `GaussianNB()` method from Scikit-learn, which works well with continuous image data, was used to implement the Naive Bayes Classifier. We trained it on 4096 pixel flattened image vectors.

Accuracy: About 87.32% accuracy was attained.

```
--- Naive Bayes Classifier ---
Accuracy : 0.8731707317073171
Precision: 0.8748053435157519
Recall   : 0.8731707317073171
F1-score : 0.8722878361871734

Classification Report:
                  precision    recall  f1-score

   glioma_tumor       0.85      0.94      0.90
       no_tumor       0.87      0.79      0.83
pituitary_tumor       0.90      0.84      0.87

       accuracy                           0.87
```

Figure 1: Naive Bayes Classifier Accuracy Metrics Results.

Three glioma tumor were incorrectly labeled as no tumor, and seven as pituitary tumor, according to the confusion matrix. This indicates how naive the model is as well as the pitfalls of assuming pixel independence.

```
Confusion Matrix:
 [[162   3   7]
 [  7  58   8]
 [ 21   6 138]]
```

Figure 2: Naive Bayes Classifier Confusion Matrix.

Decision Tree Classifier

Pixel intensity for splits was efficiently used in the implementation of the Decision Tree Classifier using `DecisionTreeClassifier` with the entropy criterion.

Accuracy: greatly increased to an accuracy of about 91.71%.

```
--- Decision Tree Classifier ---
Accuracy : 0.9170731707317074
Precision: 0.9178147554210746
Recall   : 0.9170731707317074
F1-score : 0.9169992859258719

Classification Report:
                 precision    recall  f1-score

   glioma_tumor       0.94      0.91      0.93
       no_tumor       0.89      0.97      0.93
pituitary_tumor       0.91      0.90      0.90

       accuracy                           0.92
```

Figure 3: Decision Tree Classifier Accuracy Metrics Results.

In comparison to Naive Bayes, the confusion matrix displays a significant improvement in class separation and fewer misclassifications (e.g. Just two of the 172 glioma tumor images were incorrectly classified).

```
Confusion Matrix:
 [[157   2  13]
 [  0  71   2]
 [ 10   7 148]]
```

Figure 4: Decision Tree Classifier Confusion Matrix.

The decision tree's first three levels were visualized using Scikit-learn's `plot_tree()` function, which showed good interpretability of the model by highlighting efficient splits based on pixel intensities.

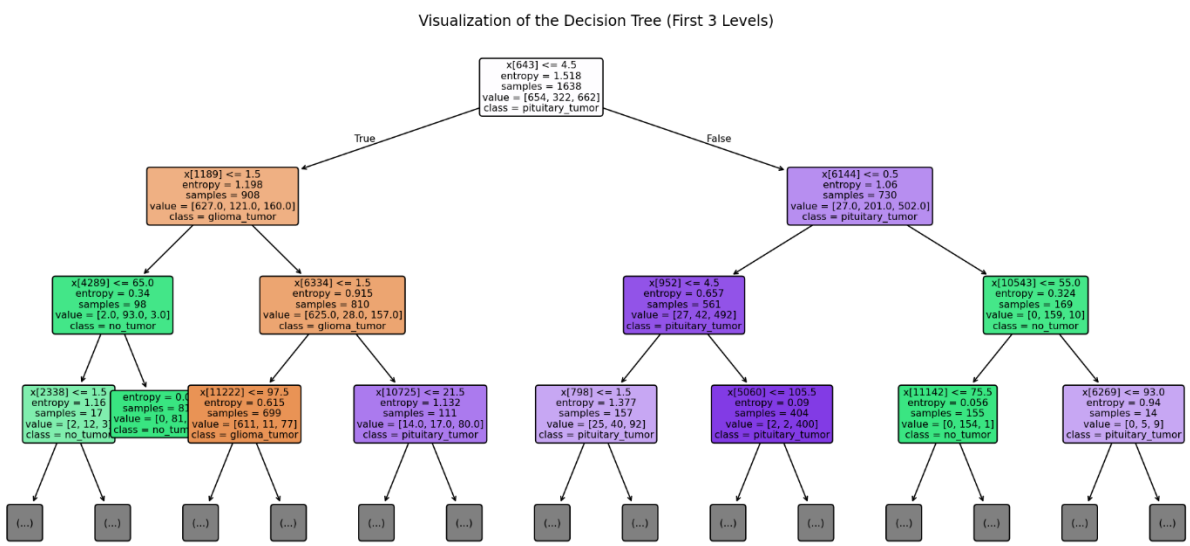

Figure 5: Decision Tree Visualization Based on Pixel Intensities.

Multi-Layer Perceptron MLP (Feedforward Neural Network)

In order to capture intricate tumor patterns, the MLP Classifier was constructed using Scikit-learn's neural network module, which has two hidden layers and was trained on scaled image data. Because it could learn non-linear relationships in high-dimensional pixel data, it performed better than the other models.

Its predictive ability makes it perfect for medical image classification tasks where accuracy is crucial, despite its computational complexity and lack of interpretability.

```
--- MLP Classifier ---
Accuracy : 0.9707317073170731
Precision: 0.9714481934314402
Recall   : 0.9707317073170731
F1-score : 0.970536656856184

Classification Report:
                  precision    recall  f1-score

   glioma_tumor       0.96      0.99      0.97
       no_tumor       1.00      0.90      0.95
pituitary_tumor       0.98      0.98      0.98

       accuracy                           0.97
```

Figure 6: MLP Classifier Accuracy Metrics Results.

Very few misclassifications were found in the MLP's confusion matrix, suggesting that the model reliably differentiated between cases of pituitary tumor, glioma, and no tumor. The model's ability to detect subtle image features was confirmed by the majority of predictions closely matching actual labels. This strengthens its dependability for practical diagnostic uses.

```
Confusion Matrix:
 [[170   0   2]
 [  5  66   2]
 [  3   0 162]]
```

Figure 7: MLP Classifier Confusion Matrix.

Model Comparison and Performance Analysis

It is clear from the analysis that Naive Bayes works best as a baseline model for short and easy image classification tasks. Its speed and simplicity of use are benefits, but when image features depend on one another, as they do in medical images, its performance deteriorates.

The interpretability and classification power of decision trees are well-balanced. They capture non-linear patterns better than Naive Bayes, but if not adjusted, they are prone to overfitting.

Because the MLP Classifier can learn intricate patterns through deep layers, it performs significantly better than both. The increased accuracy makes it the best model for accurate tumor classification in the real world, despite the fact that it necessitates more preprocessing and training time.

To sum up, The MLP Classifier is the most dependable option for brain tumor classification because it accurately captured complex image features, even though Naive Bayes and Decision Tree models offer speed and interpretability.